Unit 5
Applications and Impacts of Computing

AI · IoT · data analytics · reliable sources · assistive technology

SENSORS & ACTUATORS | SMART SYSTEMS | PRIMARY / SECONDARY / TERTIARY | HUMAN IMPACT

Computing left the lab. This unit asks what it does in a city, a farm, a hospital — and who it leaves out.

0. Three technologies that travel together

In Pakistan, students already meet computing as LMS homework, ATM cards, hospital records, and weather apps for wheat. Three ideas sit under almost every “smart” headline.

ARTIFICIAL INTELLIGENCE

Artificial = made by people. *Intelligence* = tasks that need judgement. Face unlock, YouTube recommendations, a website chatbot. AI is the “brain” of a smart system.

INTERNET OF THINGS

Physical devices on a network that collect and share data and often act without a human in the loop: watches, meters, cameras, irrigation valves.

DATA ANALYTICS

Clean, study, and visualise the flood of readings so a human (or an AI) can decide. Bar and pie charts are the last mile of analytics, not the whole job.

$$\text{IoT sensors} \xrightarrow{\text{data}} \text{Analytics} \xrightarrow{\text{patterns}} \text{AI decision} \xrightarrow{\text{command}} \text{Actuator}$$

Smart home: a temperature sensor reports $33\,^{\circ}\text{C}$; analytics confirms it is a trend, not a glitch; AI switches the inverter AC on.

1. Anatomy of an IoT system

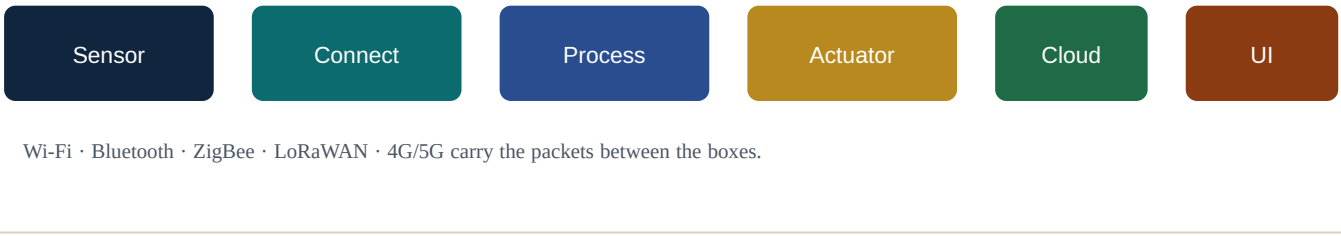


Figure 1. IoT stack. If any box is missing, the “smart” device is just a gadget.

Sensors — they listen

- Temperature, motion, smoke
- Fingerprint / camera (biometric)
- Soil moisture (farming)

A fingerprint sensor captures an image, builds a template, and compares it with stored data.

Actuators — they do

- Motor, electric switch
- Automatic door, irrigation valve

Motion detected → controller → motor opens the door. The actuator turns a bit into a movement.

Table 1. Where IoT already sits in daily life

Domain	Typical devices / jobs
Smart city	Traffic lights, street lighting, parking, waste bins
Industry	Machine health, line automation
Healthcare	Smartwatch heart-rate, remote monitoring
Education	Biometric attendance, smart classroom boards
Home	Thermostats, cameras, appliance plugs
Farming	Soil moisture + weather → smart irrigation (Sindh / Punjab examples)

1.1 Analytics pipeline

1. **Collect** from sensors, forms, logs.
2. **Store** in a file or database (Unit 4).
3. **Clean** duplicates, missing cells, wrong formats — dirty data lies.
4. **Analyse** totals, trends, outliers.
5. **Visualise** bar / pie / line so a non-programmer can see the story.

2. Information sources — not every page is a source

Table 2. Source ladder

Type	What it is	Examples	Typical risk
Primary	First-hand, original	NADRA / CNIC data, your interview, official register	Incomplete or false answers
Secondary	Someone already processed it	Textbook, PBS report, newspaper	Out of date, no citations
Tertiary	Summary of summaries	Encyclopaedia, Wikipedia overview	Oversimplified, unverified sites

A career page that lists “AI engineer, IoT technician, data analyst” is only as good as the organisation that published it. Ask: who wrote this, when, with what evidence, and what did they leave out?

3. Impacts — a balanced ledger

Table 3. Computing changes the field — both ways

Field	Gain	Cost if we are careless
Education	LMS, remote classes, instant resources	Screen fatigue, copied homework
Communication	Family and class groups	Rumour speed, cyberbullying
Business	Online shops, mobile payments	Job shift, fraud
Healthcare	Digital records, remote consults	Privacy leaks
Agriculture	Weather and irrigation advice	Cost of devices, data gaps in villages
Daily life	Maps, banking, government portals	Dependence, exclusion of the offline

4. Assistive technology — computing as inclusion

Assistive tools exist so disability does not become exclusion.

MOBILITY
Switch controls, adapted keyboards

SENSORY
Screen readers (NVDA, VoiceOver), magnification, captions

COMMUNICATION
AAC devices and symbol boards

COGNITIVE / LEARNING
Audiobooks, spell-check, text-to-speech

Related careers: assistive-software developer, AI/ML specialist, biomedical engineer, accessibility UX designer, rehabilitation technologist.

Self-check

1. In one sentence each, separate IoT, AI, and analytics.
2. Is a Wikipedia page a primary source? Why?
3. Name one sensor and one actuator in a smart irrigation sketch.

Answers. (1) IoT gathers/acts in the physical world; analytics finds patterns; AI decides. (2) No — it is tertiary (sometimes secondary); it summarises other work. (3) Soil-moisture sensor; water-valve actuator.

Recap

1. Smart loop: sensor → connect → process / analyse → AI → actuator → UI.
2. IoT domains: city, industry, health, school, home, farm.
3. Analytics is only as honest as its cleaning step.
4. Primary / secondary / tertiary — always ask who, when, evidence.
5. Every gain has a human cost; assistive tech is part of the answer, not an optional extra.

Redraw Figure 1 and label a farm example on it.